Al-Nahiyan Khushbu

Chengdu, Sichuan, China | al-nahiyan@outlook.com | github.com/docAlnahiyan | linkedin.com/in/al-nahiiyan | ORCID 0009-0001-7030-8516

SUMMARY

PhD researcher in artificial intelligence working in medical imaging and computer vision. I design and implement hybrid CNN–Transformer architectures for 2D and 3D imaging, with a focus on early Parkinson's disease recognition, and bring research (architecture design, attention mechanisms) together with practical engineering (Python, PyTorch, Docker) so that contributions are both sound and ready for real–world use.

EDUCATION

PhD in Artificial Intelligence — Southwest Jiaotong University Sep 2025 – Present
Chengdu, China

MSc in Computer Science & Technology — Southwest Jiaotong University 2022 – 2025
Chengdu, China · GPA 3.38 / 4.0

BEng in Software Engineering — Sichuan University 2017 – 2021
Chengdu, China · GPA 3.02 / 4.0

PUBLICATIONS

1. **A.-N. Khushbu**, Z. Yang, Y. Liu, and X. Zhang, "Parkinson's Disease Recognition Using Hybrid CNN–Transformer Model," *2024 4th International Conference on Industrial Automation, Robotics and Control Engineering (IARCE)*, IEEE, 2024, pp. 307–311. DOI: 10.1109/IARCE64300.2024.00064. **— Best Paper Award.**
2. **A.-N. Khushbu**, X. Zhang, A. K. M. M. Islam, Y. Liu, and Y. Pan, "HybridTransMedFormer: A Dual-Path 3D MRI Fusion Framework for Early Parkinson's Disease Recognition," *Artificial Intelligence in Medicine*. (Under review, 2026)
3. M. J. Alam and **A.-N. Khushbu**, "Catching Theft When the Records Lie: Exposure-Based Attribution of Inventory Shrinkage in Pharmaceutical Warehouses," Preprint, 2026 (not peer-reviewed).

RESEARCH & PROJECT EXPERIENCE

Master's Thesis — Hybrid Deep Learning for Early Parkinson's Disease: Bridging 2D and 3D MRI 2025

A two-tier diagnostic pipeline on structural MRI: a lightweight hybrid CNN–Transformer for 2D rapid pre-screening (80.51% accuracy) and HybridTransMedFormer, a 3D dual-path fusion architecture for confirmatory diagnosis (92.34% accuracy), projected to reduce diagnostic delay by 12–18 months for prodromal patients.

MultimodalTransformerNet — Multimodal Emotion Recognition 2025

A multimodal framework using cross-modal attention fusion to combine visual features with 39-dimensional auditory features, improving accuracy over single-input baselines.

GlobalFaceNet — Facial Expression Recognition 2025

A CNN–Transformer architecture with dual-pathway processing for fine-grained facial detail and global spatial context; 85.0% accuracy on RAF-DB with 3.2 ms inference.

Bachelor's Thesis — Bird Species Identification 2021

Image recognition with transfer learning (VGG-16, ResNet-18) on the Caltech-UCSD Birds 200 (CUB-200-2011) dataset, with preprocessing and data augmentation in PyTorch.

Other projects 2020 – 2021

Bone-fracture detection from X-rays (image processing); COVID-19 facemask detection (CNN + OpenCV, real-time); AI-driven e-commerce recommendation engine (intern project, collaborative & content-based filtering, Flask/SQL).

WORK EXPERIENCE

Software Requirements Research & Publisher — SoftFlames Ventures LLC Nov 2025 – Present

Hamtramck, Michigan, USA

Software Engineer Intern — Chengdu Suncape Data Co., Ltd Sep 2020 – Mar 2021

Chengdu, China

TECHNICAL SKILLS

AI / Deep Learning	Python, Java · PyTorch, TensorFlow, Keras, attention mechanisms
Medical / Vision	OpenCV, Grad-CAM, NiBabel, Scikit-image, 3D data augmentation
Deployment / Tools	Docker, FastAPI, Flask, SQL, Git / GitHub
Data Analysis	NumPy, Pandas, Matplotlib, Seaborn